

Dr Sophie Koudmani

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RESEARCH INTERESTS	Computational astrophysics; galaxy formation and evolution, dwarf galaxies; astrophysical black holes; multi-messenger signatures; machine-learning-accelerated simulations	
EMPLOYMENT	Royal Society University Research Fellow	Jan 2026 –
	Royal Society, UK Independent Research Fellowship Host: Centre for Astrophysics Research (CAR), University of Hertfordshire	
	Lecturer	Jan 2025 –
	University of Hertfordshire, Hatfield, UK Proleptic Appointment Department of Physics, Astronomy and Mathematics	
	1851 Royal Commission Fellow	Oct 2024 – Dec 2025
	The Royal Commission for the Exhibition of 1851, UK Independent Research Fellowship Host: CAR, University of Hertfordshire	
	Junior Research Fellow	Oct 2021 – Sep 2025
	St Catharine's College, University of Cambridge, UK Independent Research Fellowship Host: Institute of Astronomy (IoA), University of Cambridge	
	Flatiron Research Fellow	Aug 2022 – Dec 2023
	Center for Computational Astrophysics (CCA), Flatiron Institute, USA Independent Research Fellowship Member of Galaxy Formation Group led by Dr Rachel Somerville	
	UROP Summer Research Scholar	Aug 2016 – Sep 2016
	Imperial Centre for Inference and Cosmology (ICIC), Imperial College London, UK Project Title: 'Inferring Ionization Maps from 21cm Maps of the Epoch of Reionization' Supervisor: Dr Jonathan Pritchard	
EDUCATION	PhD in Astronomy	2017-21
	IoA and Churchill College, University of Cambridge, UK Thesis: 'Black Hole Feedback in New Regimes: Modelling Dwarf Galaxies with Active Galactic Nuclei' Supervisors: Prof. Debora Sijacki and Prof. Martin Haehnelt	
	MPhys Physics (1st Class)	2013-17
	New College, University of Oxford, UK MPhys Project: 'The HI Mass Function in the Horizon-AGN Simulation' (1st Class) Supervisors: Prof. Adrianne Slyz and Prof. Julien Devriendt	
HONOURS & AWARDS	University Research Fellowship, Royal Society	2025
	1851 Research Fellowship, Royal Commission for the Exhibition of 1851	2024
	Flatiron Research Fellowship (<i>deferred from 2021</i>), CCA, Flatiron Institute	2022
	Junior Research Fellowship, St Catharine's College, Cambridge	2021
	Paul Murdin Prize (<i>best paper by an IoA PhD student</i>), IoA, Cambridge	2019
	Fully-funded PhD Studentship, STFC	2017-21
	UROP Research Scholarship, Imperial College London	2016
	Commendation for Laboratory Practical Work, Oxford Physics Department	2016
	Scholarship Prize, New College, Oxford	2015-17
	Physics Abitur Prize, German Physical Society (DPG)	2013
	Mathematics Abitur Prize, German Mathematical Society (DMV)	2013

SELECTED GRANTS & FUNDING	JWST, Cycle 4 (6.7 hrs), STScI, Co-I, PI: Scholtz	2025
	Kavli Workshop ‘Massive Black Holes Across Cosmic Time’, Kavli Foundation (Co-Leader w/ Übler)	2025
	JWST Large Program, Cycle 3 (137.82 hrs), STScI, Co-I, PI: Übler	2024
	CCA Workshop ‘ Black Holes on Broadway ’, Simons Foundation (Co-Leader w/ Rennehan)	2023
	DiRAC Computing Grant (32.77 MCPUh), STFC, Co-I & Sub-Project Leader , PI: Sijacki	2023
	DiRAC Computing Grant (2.99 MCPUh), STFC, PI , [DiRAC Science Highlight]	2021
	DiRAC Computing Grant (35.72 MCPUh), STFC, Co-I & Sub-Project Leader , PI: Sijacki	2020
	Conference Travel Support (\$900), North American ALMA Science Center	2020
	Conference Travel Grants (£700), Churchill College, Cambridge	2019, 2020
	DiRAC Computing Grant (28.47 MCPUh), STFC, Co-I & Sub-Project Leader , PI: Sijacki	2018
	Undergraduate Research Bursary (£1200), Royal Astronomical Society	2016

SELECTED TALKS	Selected Conference Talks (30 in total, 12 invited):	
	<i>Invited Review</i> , LtU Annual Collaboration Meeting, NYC, USA	Sep 2025
	<i>Invited</i> , ‘Advancing massive black hole modelling across galactic scales’, Helsinki, Finland	Jul 2025
	<i>Invited Review</i> , EAS SS36 ‘From Giants to Dwarfs’, Cork, Ireland	Jun 2025
	<i>Invited Review</i> , MIAPbP Workshop ‘Unveiling a Universe of Black Holes’, Garching, Germany	Mar 2025
	<i>Invited</i> , Sesto Workshop ‘The Growth of Galaxies in the Early Universe’, Sesto, Italy	Jan 2025
	<i>Invited</i> , Ringberg Workshop on ‘Computational Galaxy Formation’, Ringberg, Germany	Oct 2024
	<i>Invited</i> , YAGN24 Workshop, Università degli Studi dell’Insubria-Como, Italy	Sep 2024
	<i>Invited</i> , Santa Cruz Workshop on Galaxy Formation, Santa Cruz, USA	Jul/Aug 2024
	<i>Contributed</i> , Massive Black Holes in the First Billion Years, UCC, Cork, Ireland	Apr 2024
	<i>Contributed</i> , EAS S12 ‘Breaking down the AGN frontiers’, Krakow, Poland	Jul 2023
	<i>Invited</i> , Learning the Universe Collaboration Meeting, Paris, France	Feb 2023
	<i>Invited Review</i> , The Epoch of Galaxy Quenching, Cambridge, UK	Sep 2022
	<i>Invited</i> , EAS S6 ‘Properties & impact of large-scale multiphase AGN outflows’, Valencia, Spain	Jun 2022
	<i>Invited</i> , Ringberg Workshop on ‘Breakthroughs in Galaxy Formation’, Ringberg, Germany	Apr 2022
	<i>Contributed</i> , SAZERAC SIP: ‘Models and Simulations of High-Redshift Galaxies’ [virtual]	Oct 2021
	<i>Contributed</i> , ‘Galaxy Quenching and Transformation Throughout Cosmic Time’, Aspen, USA	Feb 2020
	<i>Contributed</i> , ‘Small Galaxies, Cosmic Questions’, Durham, UK	Jul 2019

Selected *Invited* Departmental Seminars & Colloquia (35 in total):

Kavli New Frontiers Lecture, University of Cambridge, UK	Mar 2025
Astronomy Colloquium, Durham University, UK	Nov 2024
Physics Colloquium, Maynooth University, Ireland	Oct 2024
Cosmology/Extragalactic Seminar, UCL, London, UK	May 2024
CCA Lunch Seminar, Flatiron Institute, NYC, USA	Apr 2024
CTC Theory Seminar, University of Maryland, USA	Apr 2024
Hernquist Galaxy Formation Group Meeting, CfA, Harvard, USA	Dec 2023
Physics Colloquium, Montana State University, USA	Sep 2023
CIERA Theory Seminar, Northwestern University, USA	May 2023
Galread Seminar, Princeton University, USA	Oct 2022
Fellows’ and Graduates’ Seminar, St Catharine’s College, Cambridge, UK	Mar 2022

STUDENT SUPERVISION	Graduate Students:	
	<i>PhD Project</i> , Kasar Profit, CAR, University of Hertfordshire	2025-29
	‘Cosmic Collisions: Predicting Gravitational Wave and EM Signals from Supermassive Black Hole Mergers’	
	<i>PhD Project</i> , Giulia Ortame, KICC, University of Cambridge	2024-28
	‘Seeds to supermassive: modelling cross-scale black hole dynamics and growth in the multi-messenger era’	
	<i>MPhil Project</i> , Brian Jiang, KICC, University of Cambridge	2023-24
	‘Early black hole growth: observational evidence for super-Eddington?’	

Undergraduate Students:

<i>MSci (Part III) Project</i> , Aineias Arango, IoA, University of Cambridge	2024-25
‘A neural galaxy: developing machine learning-enhanced simulations of the black hole – galaxy connection’	
<i>Summer & MAsT (Part III) Project</i> , Giulia Ortame, IoA, University of Cambridge	2023-24
‘Unveiling Black Hole Accretion in Dwarf Galaxies’	
<i>MSci (Part III) Project</i> , Arjun Swarup, IoA, University of Cambridge	2023-24
‘The complex interplay of stellar and AGN feedback in simulated galaxies’	
<i>Summer Project</i> , Kasar Profit, CCA, Flatiron Institute	2023
‘Merging Supermassive Black Holes’	
<i>MSci (Part III) Project</i> , Guy Uong, IoA, University of Cambridge	2021-22
‘Dynamical Friction on Supermassive Black Holes in Galaxy Formation Simulations’	

TEACHING

Lectures:

Advanced Research Topics, MSc Data Science, University of Hertfordshire	2025
STFC Introduction to Astronomy Summer School, University College London	2025
Astrophysical Black Holes (<i>Substitute Lecturer</i>), 4th Year Astrophysics, University of Cambridge	2022

Tutorials & Small-Group Teaching:

Advanced Research Topics (<i>Computing Tutorials</i>), MSc Data Science, University of Hertfordshire	2025
MSc Data Science Projects (<i>Group Supervisions</i>), University of Hertfordshire	2024-25
Astrophysical Black Holes (<i>Drop-in Sessions</i>), 4th Year Astrophysics, University of Cambridge	2019,21
Astrophysical Fluid Dynamics (<i>Supervisions</i>), 3rd Year Astrophysics, University of Cambridge	2019
Mathematics (<i>Supervisions</i>), 1st Year Natural Sciences, University of Cambridge	2017-18

Examinations:

Coursework in Advanced Research Topics, MSc Data Science, University of Hertfordshire	2025
MSc Data Science Projects, University of Hertfordshire	2024-25
IoA Internal PhD Thesis Review, University of Cambridge	2022,24,25
Standards Assessor, Astrophysical Black Holes, 4th year Astrophysics, University of Cambridge	2022
Marks Checker, Astrophysical Black Holes, 4th year Astrophysics, University of Cambridge	2019,21,22

SERVICE

Conference Organiser:

Co-Organiser: 'Massive Black Hole Spin', Higgs Centre Workshop, University of Edinburgh	2026
LOC Co-Chair: 'Massive Black Holes Across Cosmic Time' conference hosted by KICC	2025
Co-Chair: Kavli Focus Meeting 'Dark Matter in Astrophysical Laboratories'	2024
SOC/LOC Co-Chair: 'Black Holes on Broadway' conference hosted by Flatiron Institute	2023
Co-Organiser: NAM Session 'Linking simulations to observations to understand galaxy evolution'	2022
Co-Organiser: Kavli Focus Meeting 'Feedback in and around Galaxies'	2021
Co-Organiser: Conference for Undergraduate Women in Physics hosted by Oxford University	2016,17

Peer Reviewer:

Reviewer for <i>NASA</i> Observational Programs	2025-
Reviewer for <i>UKRI</i> Fellowship Awards	2025-
Reviewer for <i>SNSF-ISF</i> Research Grants	2025-
Reviewer for <i>Nature Astronomy</i>	2025-
Reviewer for <i>ERC</i> Starting Grant	2024-
Book Reviewer for <i>The Observatory</i>	2023-
Reviewer for <i>Open Journal of Astrophysics</i> (OJAp)	2023-
Reviewer for <i>The Astrophysical Journal</i> (ApJ)	2022-
Reviewer for <i>Monthly Notices of the Royal Astronomical Society</i> (MNRAS)	2022-

Equality, Diversity and Inclusion:

Organiser of STEMPoint coding workshops for Year 8/9 girls from Hertfordshire schools	2025
Mentor for the National Society of Black Physicists (NSBP) Summer Research Programme	2023
IoA Self Assessment Team: Juno/Athena SWAN Gender Equality Awards	2020-22
IoA Equality, Diversity & Inclusion Committee	2018-22
Oxford Women in Physics Society: Undergraduate Representative & Mentoring Coordinator	2015-17
Oxford University Counselling Service: Trained Peer Supporter for New College	2014-17
New College JCR Committee: International Student Representative	2014-15

Outreach and Public Engagement:

Organiser of Kavli Public Lecture	2025
Developer of 'Cosmic Coders' Python Course for high school students	2025
Interview: ' <i>Webb has shown us they are clearly wrong</i> ', Live Science	2024
Feature: ' <i>The astrophysicist unravelling the origins of supermassive black holes</i> ', New Scientist	2024
Invited Speaker at New Scientist Live, ExCeL London	2024
Interview: ' <i>Advances in our understanding of the Universe since 1473</i> ', St Catharine's College	2023
4 Outreach Talks at IoA and Churchill College	2018-22
Volunteer at Cambridge Science Festival, IoA	2018-19
Course Instructor at 'Greenlight for Girls' STEM Initiative	2017

Other Departmental Activities:

Computational Astrophysics Group Meeting (Co-Organiser & -Initiator), Hertfordshire	2024-
Galaxy Formation Group Meeting (Co-Chair), CCA, NYC	2022-23
IoA Computer Users Committee, Cambridge	2018-22
Black Hole Physics Student Discussion Group (Organiser & Initiator), Cambridge	2019-21

COMPUTING SKILLS *Programming Languages:* Python, C, BASH scripting
Tools/Packages: numpy, scipy, matplotlib, Git, MPI

PROFESSIONAL International Collaborations:

MEMBERSHIPS	NewAthena, Member of ‘Galaxies and Supermassive Black Holes’ Working Group	2025-
	LSST UK, Rubin Observatory, Junior Associate	2025-
	BlackTHUNDER, JWST Large Program, Theoretical interpretation of observations with AESOPICA	2024-
	‘Learning the Universe’ Collaboration, Co-Chair of Black Hole Working Group	2022-
	LISA Consortium, Member of Astrophysics Working Group	2021-

Learned Societies:

American Astronomical Society	2020-23
Royal Astronomical Society	2019-
Institute of Physics	2019-
Cambridge Philosophical Society	2019-
German Physical Society (DPG)	2013-

REFERENCES **Dr Rachel Somerville**
Galaxy Formation Group Leader, Flatiron Institute, USA
Email: rsomerville@flatironinstitute.org

Prof. Debora Sijacki
Professor of Astrophysics and Cosmology, University of Cambridge, UK
Email: deboras@ast.cam.ac.uk

Prof. Adrianne Slyz
Associate Professor of Astrophysics, University of Oxford, UK
Email: adrianne.slyz@physics.ox.ac.uk

PUBLICATIONS *As of Oct 2025:*
17 papers accepted/submitted with over 450 citations (over 260 citations in 2025).
5 first-author papers accepted with over 230 citations (over 90 citations in 2025).
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[\[ADS link for first-author publications\]](#)

[\[ADS link for all publications\]](#)